



Influence of Fe_2O_3 on non-isothermal crystallization kinetics and microstructure of lithium titanium phosphate glass-ceramics

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ABSTRACT

The crystallization behavior and microstructure of $\text{Li}_2\text{O}-\text{TiO}_2-\text{P}_2\text{O}_5-\text{Fe}_2\text{O}_3$ glass-ceramic were investigated by means of differential thermal analysis (DTA), X-ray diffraction (XRD) and scanning electron microscopy (SEM). Crystallization kinetic parameters including the activation energy for crystallization (E_c) and the Avrami exponent (n) were also measured. The activation energy for crystallization was found to be 656 kJ/mol and 295 kJ/mol, respectively, in the Fe_2O_3 -free and 5 weight ratio Fe_2O_3 containing glasses. The FTIR spectra revealed a strong band due to the influence of Fe_2O_3 content on different phosphate groups. Impedance spectroscopy studies suggest that the ionic conductivity of the glass-ceramics containing the optimized Fe_2O_3 content has been improved as compared with the Fe_2O_3 free one. According to the obtained results, $\text{LiTi}_2(\text{PO}_4)_3$ precipitated as near spherical and pyramid-like crystals.

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1. Introduction

Phosphate glasses exhibit attractive properties such as low glass transitions and melting temperatures, high thermal expansion coefficient, biocompatibility, and high refractive indices. These specifications make them suitable for many applications such as photonics, fast ion conductors [1–4], glass to metal seals [5–7], low temperature enamels [7], NH_3 gas adsorption [8], and biomedical engineering [9]. Recently, lithium phosphate glass-ceramic electrolytes have been widely studied due to various technological advantages such as great varieties of composition, absence of grain boundaries, ease of thin film fabrication, elimination of porosity, molding in desired shapes and long range disorder leading to high ionic conductivity [1,10–13].

The glass forming ability of the $\text{Li}_2\text{O}-\text{TiO}_2-\text{P}_2\text{O}_5$ system glasses was investigated by Kishioka. He has identified the restricted glass-forming regions of this system [14]. Glass-ceramics belonging to the $\text{Li}_2\text{O}-\text{TiO}_2-\text{P}_2\text{O}_5$ system exhibit superior properties such as ionic conductivity [2] and ion-exchange capacity [15] due to the formation of an alkali or alkaline-earth titanium phosphate phase which is called NASICON [4, 15–17]. NASICON crystals with the formula of $\text{LiTi}_2(\text{PO}_4)_3$, built up by corner-sharing of TiO_6 octahedron and PO_4 tetrahedron, form a class of super ionic conductors [4,18–20]. In the lithium ion conductive glass-ceramics, the crystalline and the residual glass phases are both

responsible for ionic conductivity. However, the role of the conductive crystalline phase such as NASICON is more highlighted in this phenomenon. In the present work, the role of Fe_2O_3 on nucleation and crystallization behavior, activation energy for crystallization and the resulted microstructure of $\text{Li}_2\text{O}-\text{TiO}_2-\text{P}_2\text{O}_5$ have been investigated. According to the previous studies [20], lithium titanium phosphate NASICON structures fabricated through classic solid state sintering showed the improved ionic conductivity in the presence of Fe_2O_3 . In fact, in these structures, replacement of Ti^{4+} ions by Fe^{3+} ions led to excess amount of Li^+ ions in the structure. Due to various valences of Ti^{4+} and Fe^{3+} ions, by replacement of Fe^{3+} instead of Ti^{4+} , Li^+ ions contribute into the NASICON structure to keep charge balance. These excess amounts of Li^+ are transferred from the residual glass matrix into the NASICON structure. As a result, by increasing the concentration of mobile Li^+ ions, ionic conductivity would be increased. However, the effect of Fe_2O_3 on the crystallization behavior, structure and ionic conductivity of the prototype glass-ceramic system has not yet been investigated.

2. Experimental procedure

2.1. Sample preparation

Table 1 shows the chemical compositions of the prepared glasses. The starting materials were reagent grade chemicals of lithium carbonate (Merck, 5670), titanium oxide (Merck, 808), iron oxide (Merck, 3924) and phosphorus oxide (Merck, 540). The homogenized batch

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Table 1

The nominal chemical compositions of the studied glasses (weight ratio).

Sample	Li ₂ O	TiO ₂	P ₂ O ₅	Fe ₂ O ₃
G _{base}	17.7	26.3	56	0
G _{Fe-2.5}	17.7	26.3	56	2.5
G _{Fe-5}	17.7	26.3	56	5
G _{Fe-7.5}	17.7	26.3	56	7.5

mixtures of these materials were melted in alumina crucibles at 1350 °C in an electrical furnace for 1 h. The molten glasses were then casted in a preheated graphite mold. The obtained glass specimens were annealed under their glass transition temperatures, i.e. at 400 °C for 1 h and allowed to cool naturally down in the furnace. After annealing, the glass plates were cut into a number of bars with dimensions of 10 × 10 × 5 mm.

2.2. Characterization methods

The crystallization behavior of the glasses was monitored by differential thermal analyzers (Polymer Laboratories, DTA-1640) in air atmosphere and at the heating rate of 10 °C/min. α -Al₂O₃ powder was used as the reference material in these experiments. This test was performed for two different glass particle sizes of 0.5–0.6 mm and <56 μ m, to evaluate the crystallization mechanism of each glass, which was derived by considering the difference of crystallization peak temperatures ($T_{c(\text{coarse})} - T_{c(\text{fine})}$) of glasses with different particle sizes.

In order to investigate the role of Fe₂O₃ in the related glass structure, Fourier transform infrared spectroscopy (FTIR) (using the Thermo

Table 2

Characteristic temperatures and glass-forming measurements extracted from the DTA thermographs.

Sample	T _g (± 0.3 , °C)	T _d ^a (± 0.3 , °C)	T _c (± 0.3 , °C)	ΔT	S
G _{base}	486	496	636	150	8.91
G _{Fe-2.5}	463	490	620	157	11.68
G _{Fe-5}	469	496	607	138	8.92
G _{Fe-7.5}	469	491	606	137	7.00

^a Dilatometric softening point temperature.

Nicolet Nexus 870 FTIR spectrophotometer) was used in the wave number range of 400–2800 cm⁻¹. Measurements were carried out immediately after preparing the desired discs at room temperature.

The crystallization kinetic of the parent glass and the optimized one was also surveyed by DTA examination at various heating rates of 5, 10, 15 and 20 °C/min. The activation energy for crystallization was calculated according to the Marotta, the Augis–Bennett and the modified Kissinger methods. The Avrami exponent was also measured through the Ozawa and Marotta methods [21–27].

Crystallinity of the heat treated specimens was identified by using an X-ray diffractometer (XRD) (Stoe, D-64295). The XRD patterns were recorded within the Bragg angle range of 10–80° at the step size of 0.02.

Field emission scanning electron microscope (FESEM) (MIRA\\TESCAN) was used to identify the microstructure of the heat treated samples. The samples were polished and then chemically etched by immersing in the dilute solution (5 wt.%) of HCl for 60 s.

For the electrical conductivity measurements, Au electrodes were sputtered on both sides of the samples. The impedance measurements were performed in air using a potentiostat (Autolab, PGSTAT 302 N), and data acquisition was made over the frequency ranges from 10 Hz to 1 MHz applying the voltage of 100 mV.

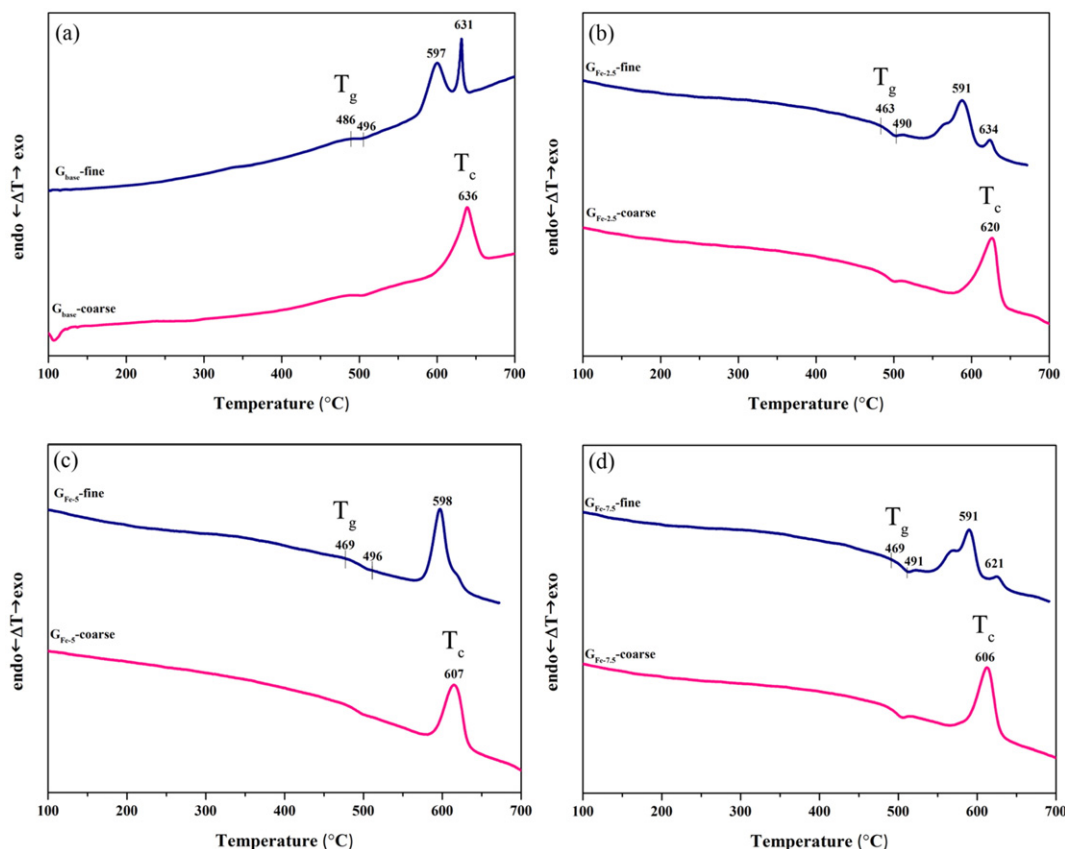


Fig. 1. DTA thermograph of 0.5–0.6 mm (coarse) and <56 μ m (fine) glass particles at the heating rate of 10 °C/min: (a) G_{base} (b) G_{Fe-2.5} (c) G_{Fe-5} and (d) G_{Fe-7.5}.

3. Theoretical background

3.1. Thermal stability of glass

A parameter which is usually employed to estimate the glass stability from the point of view of glass resistance against devitrification, is defined as ΔT_c [28], which is equal to:

$$\Delta T_c = T_c - T_g \quad (1)$$

where T_c and T_g are the crystallization and the glass transformation temperatures of glass, respectively. A greater ΔT_c means a more stable glass against crystallization.

Saad and Poulin [29] have recommended another parameter for the thermal stability of a glass:

$$S = (T_c - T_x) / (T_x - T_g) / T_g \quad (2)$$

Here S reflects the glass forming ability during cooling of the molten glass and T_x is the crystallization onset temperature. A greater S parameter reflects higher stability against crystallization.

3.2. Kinetic of crystallization

The theoretical basis for interpreting the crystallization mechanism of amorphous materials is provided by the general Johnson–Mehl–Avrami (JMA) equation [30,31]. According to this Eq., a material fraction (x) that crystallized under isothermal condition and soaking time t is given by Eq. (3):

$$x = 1 - \exp[-(kt)^n] \quad (3)$$

where n is the Avrami exponent that gives some details about the dimensionality of crystal growth [30,31] and k is actually a measure of the crystallization rate. The parameter k usually represents the Arrhenius temperature dependence (Eq. (4)):

$$k = k_0 \exp\left(-\frac{E_c}{RT}\right) \quad (4)$$

where k_0 is the frequency factor indicating the number of attempts to overcome the energy barrier and E_c is the activation energy for crystallization. Based on the JMA equation and from the heating rate dependence of the onset (T_x) and crystallization peak temperatures (T_c), several theoretical relations for the non-isothermal conditions and a

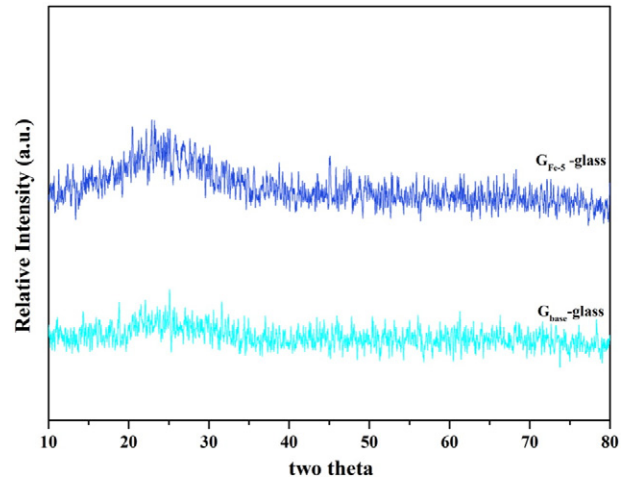


Fig. 3. The XRD patterns of the glass samples after annealing at 400 °C.

lot of approximations [21–23] have been developed to calculate the amounts of E_c and n . One of them is the relation which has been derived by Marotta [24] and is given as Eq. (5):

$$\ln \alpha = -\frac{E_c}{RT_c} + \text{Const.} \quad (5)$$

The modified Kissinger relation is another common model which is used to evaluate the E_c . This relation is given as [22]:

$$\ln\left(\frac{\alpha}{T_c^2}\right) = -\frac{E_c}{RT_c} + \text{Const.} \quad (6)$$

Augis and Bennett [23] developed an approximation method to determine the E_c according to the following relation:

$$\ln\left(\frac{\alpha}{T_c^2}\right) = -\frac{E_c}{RT_c} + \text{Const.} \quad (7)$$

In the present work the as-mentioned relations were used to determine the activation energy of crystallization of the glasses.

The Avrami exponent (n) can be extracted from the Marotta and the Ozawa equations, respectively [24–27]:

$$\ln \Delta T = -\frac{nE_c}{RT_c} + \text{Const.} \quad (8)$$

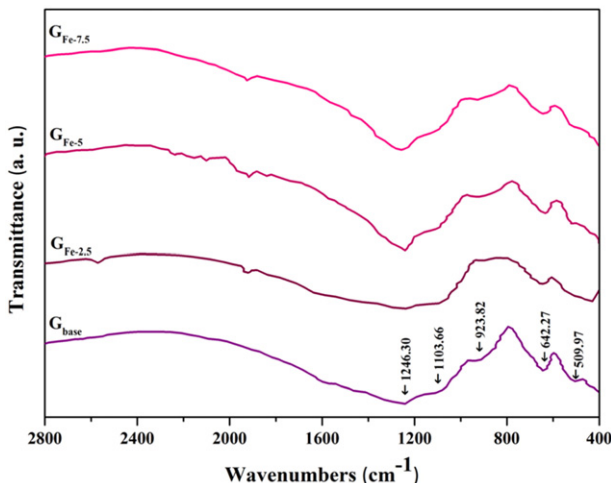


Fig. 2. FTIR transmission spectra of the studied glasses.

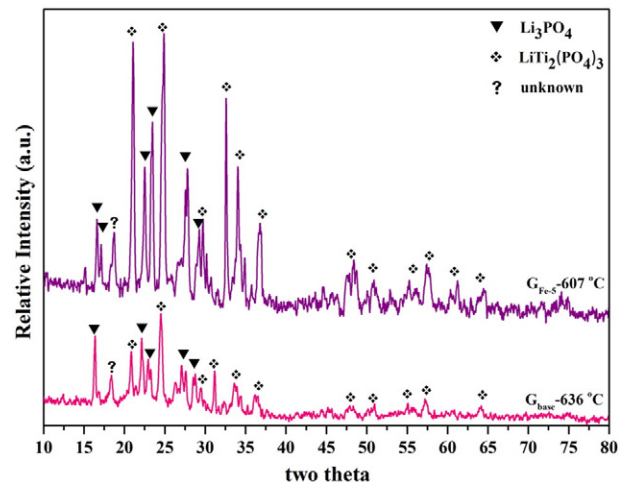


Fig. 4. The XRD patterns of the G_{base} heat treated at 636 °C for 5 min and G_{Fe-5} heat treated at 607 °C for 5 min.

$$\ln[-\ln(1-x)] = -n \ln \alpha + \text{Const.} \quad (9)$$

In these equations, α , R , T_c , ΔT , x and E_c are referred to the heating rate, gas constant, crystallization temperature, and deviation from the baseline, volume fraction of crystallization and the activation energy for crystallization, respectively.

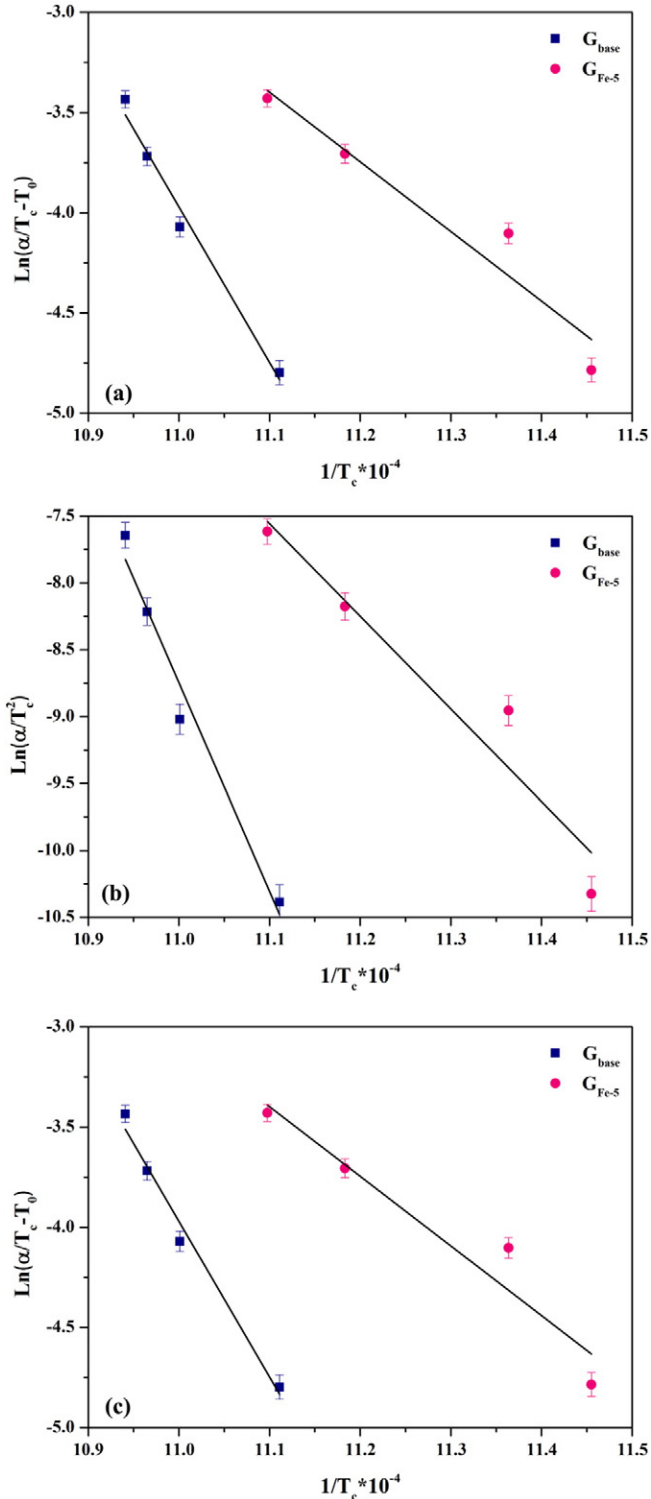


Fig. 5. Representative activation energy plots, derived by linear regression for crystallization of G_{base} and $G_{\text{Fe-5}}$ glasses: (a) the Marotta (b) the modified Kissinger (c) the Augis–Bennett methods.

4. Results and discussion

4.1. Differential thermal analysis

The DTA curves of the fine and coarse particles of G_{base} and Fe bearing glasses are shown in Fig. 1. As it can be seen there are two crystallization peaks in each fine glass powder trace, while only one crystallization peak is detectable in the coarse glass one's traces. Therefore, it seems that the crystallization behavior of the studied glasses is definitely dependent on the particle size. Among the studied glasses, the shape and the crystallization peak position of the $G_{\text{Fe-5}}$ do not change so much by changing its particle size. Apparently, the crystallization mechanism has been changed from surface to bulk in the case of glass $G_{\text{Fe-5}}$. Therefore, the role of Fe_2O_3 was studied in this glass by calculation of the activation energy for crystallization and the Avrami exponent.

The characteristic temperatures of the glass samples as well as their calculated criteria for glass forming ability have been collected in Table 2. It is realized that contrary to the nearly same softening point temperature, the crystallization peak temperatures of glasses are gradually decreased with increasing the content of Fe_2O_3 . Evaluation of the glass forming ability criteria (ΔT_c and S parameters) revealed the most glass forming ability of the composition $G_{\text{Fe-2.5}}$ compared to the other glass compositions. To understand better the role of Fe_2O_3 in the glass forming ability, the FTIR spectra of the studied glasses were also considered.

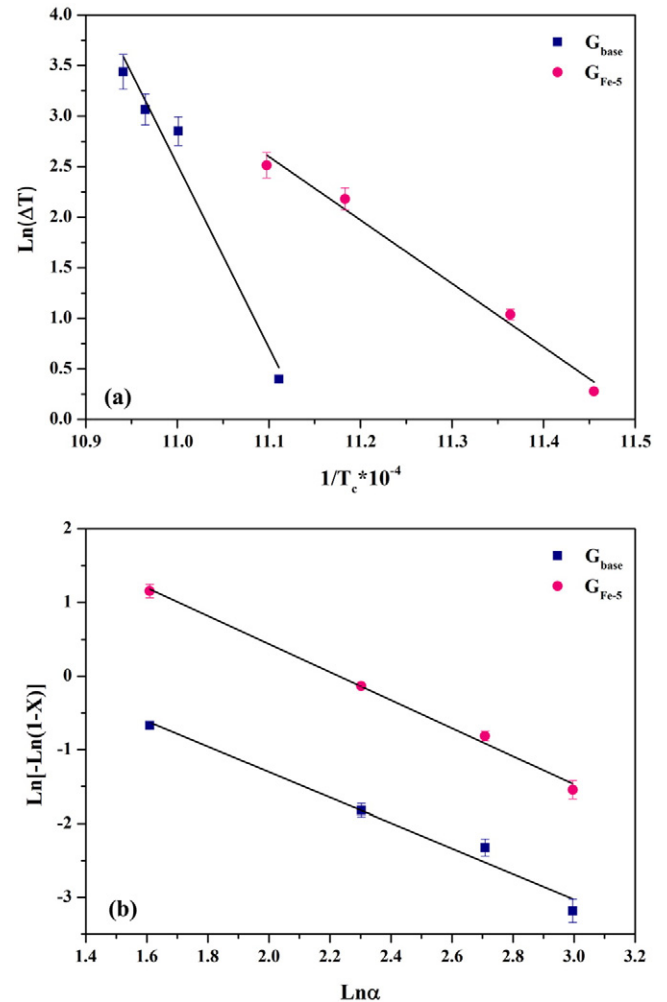


Fig. 6. Experimental plots of calculation of the Avrami exponents for G_{base} and $G_{\text{Fe-5}}$ glasses according to the (a) Marotta and the (b) Ozawa methods.

Table 3Crystallization kinetic parameters of the glasses G_{base} and $G_{\text{Fe-5}}$.

Sample	T_c (± 0.3 , °C) (at heating rates of 5, 10, 15, 20 °C/min)	E_c (Marotta) (kJ/mol)	E_c (modified Kissinger) (kJ/mol)	E_c (Augis–Bennett) (kJ/mol)	n (Marotta)	n (Ozawa)
G_{base}	620, 636, 639, 643	656 ($r^2 = -0.988$)	641 ($r = -0.987$)	636 ($r = -0.992$)	0.60 ($r = -0.985$)	0.66 ($r = -0.990$)
$G_{\text{Fe-5}}$	600, 607, 615, 624	295 ($r = -0.967$)	288 ($r = -0.965$)	289 ($r = -0.966$)	1.77 ($r = -0.994$)	1.81 ($r = -0.998$)

^a Pearson's correlation coefficient.

4.2. Infrared spectroscopy

FTIR spectra of the glasses in the wave number range of 400–2800 cm^{-1} are shown in Fig. 2. The studied glasses show nearly the similar IR spectra. The band located at 509 cm^{-1} may be attributed to the deformation modes of phosphate tetrahedral and the vibrations of anion oxygen polyhedron [32]. The band located at about 650 cm^{-1} is assigned to bond bending vibration of $\text{O}=\text{P}-\text{O}$ [33]. This band can also be related to the vibrations of $\text{Ti}-\text{O}$ bonds in the deformed octahedral of TiO_6 [34–36]. The bands located at 930 and 1250 cm^{-1} are caused by the symmetric and asymmetric stretching vibrations of

$\text{P}-\text{O}-\text{P}$ linkages in Q^2 units [37–40], respectively, indicating that the Q^2 units exist in the glass structure. In the case of $G_{\text{Fe-5}}$ and $G_{\text{Fe-7.5}}$ a strong band near 1200–1280 cm^{-1} can be observed which is assigned to the asymmetric stretching mode of two non-bridging oxygen atoms bonded to phosphorus atoms, the $\text{O}-\text{P}-\text{O}$ or $(\text{PO}_2)_{\text{as}}$ units in the phosphate tetrahedral [41,42]. It means that by increasing the Fe_2O_3 content, non-bridging oxygens in the glass structure are increased. This phenomenon confirms that Fe_2O_3 acts as a modifier in the structure and weakens the structural bonds. According to Table 2, the addition of 2.5 weight ratio of Fe_2O_3 decreases T_g which means a decrease of glass viscosity. This result confirms the modifying role of 2.5 weight ratio of

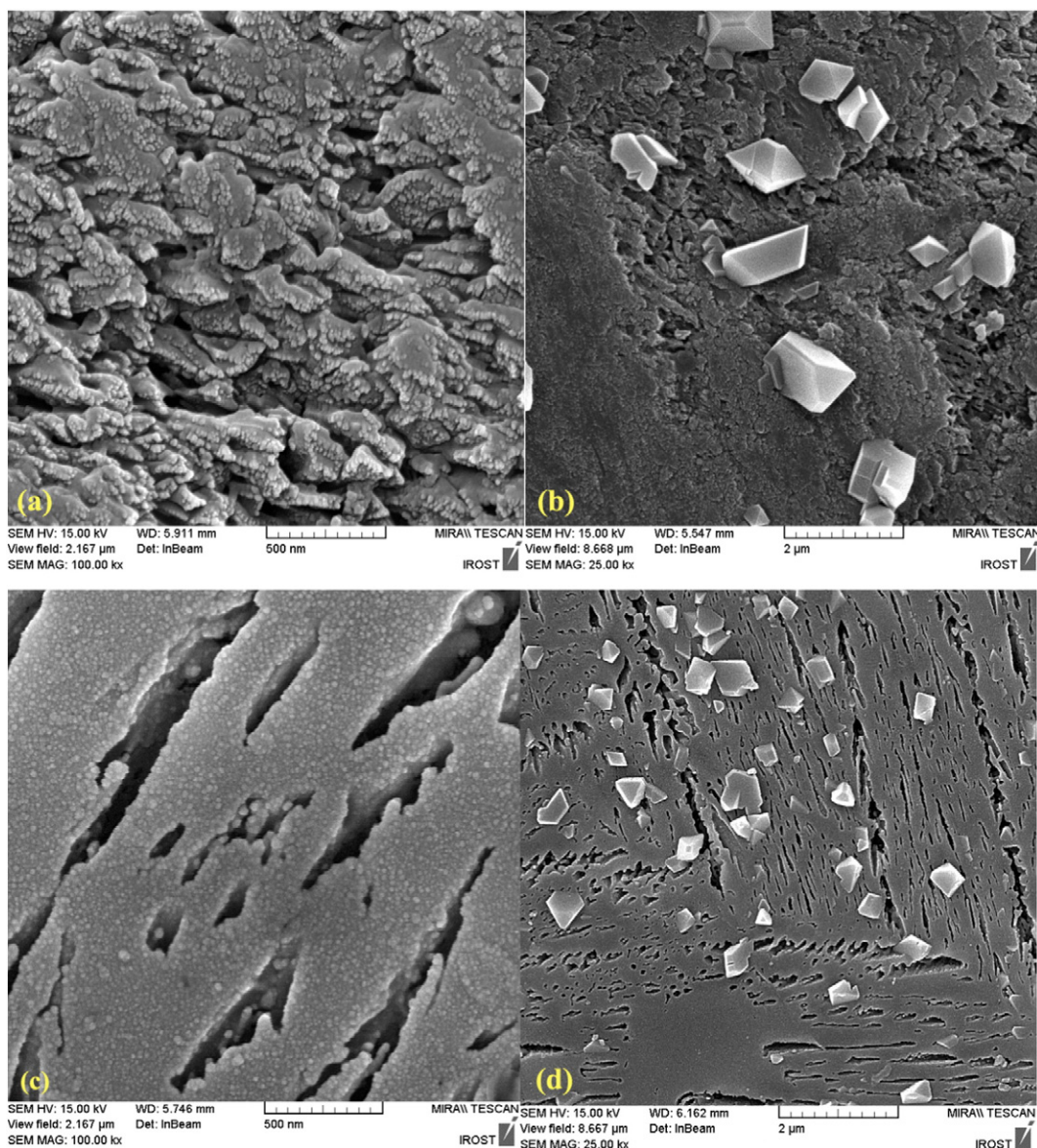


Fig. 7. SEM micrographs of the crystallized specimens: G_{base} heat treated (636 °C) at (a) 100,000 magnification and (b) 25,000 magnification, and $G_{\text{Fe-5}}$ heat treated (607 °C) at (c) 100,000 magnification and (d) 25,000 magnification.

Fe_2O_3 extracted from the FTIR spectra. Although further increase of Fe_2O_3 does not follow the similar trend, however the T_g of the glasses $G_{\text{Fe-5}}$ and $G_{\text{Fe-7.5}}$ is less than that of the G_{base} .

4.3. X-ray diffraction analysis

Fig. 3 represents the XRD patterns of the annealed glass specimens. The absence of the Bragg peaks confirms the amorphous inheritance of the fabricated glasses. It is worthy to note that when uncontrolled crystallization occurs during glass fabrication, further investigation about glass crystallization behavior and its mechanism would be meaningless. Fig. 4 shows the XRD patterns of the coarse particles of the specimens G_{base} and $G_{\text{Fe-5}}$ after 5 min heat treatment at their crystallization peak temperatures. In both crystallized samples, lithium titanium phosphate ($\text{LiTi}_2(\text{PO}_4)_3$) and lithium orthophosphate (Li_3PO_4) are present as the major and minor crystalline phases, respectively. Since no Fe bearing

crystalline phases were detected in the specimen $G_{\text{Fe-5}}$, it can be deduced that Fe^{3+} ions have been mainly remained in the residual glass matrix.

A comparison between the intensity of crystalline phases in both specimens reveals that $G_{\text{Fe-5}}$ has been more crystallized than G_{base} . It is implied that in the case of both crystalline phases, the strong lines of each crystalline phase are more intensified in the sample $G_{\text{Fe-5}}$ rather than G_{base} . Furthermore, a comparison between the crystalline phases intensity in each glass-ceramic indicates the improved and dominant crystallization of the NASICON phase in the $G_{\text{Fe-5}}$. This result suggests that the crystallization peak of the glass $G_{\text{Fe-5}}$ (Fig. 1c) is related to the NASICON phase formation. By considering this behavior and lower crystallization temperature of $G_{\text{Fe-5}}$, it can be deduced that $G_{\text{Fe-5}}$ has been crystallized easier compared to the G_{base} . Regarding the improved crystallization behavior and bulk crystallization tendency, $G_{\text{Fe-5}}$ was considered as the optimized composition and chosen for further examination.

4.4. Kinetics of crystallization

The crystallization mechanism of the glasses G_{base} and $G_{\text{Fe-5}}$ was evaluated by determining their activation energy for crystallization and their Avrami exponent.

Fig. 5a, b and c shows the Marotta, the modified Kissinger and the Aguis–Bennett-based plots, by which the activation energy for crystallization has been extracted. In this way different heating rates (α) of 5, 10, 15 and 20 °C/min were used. The activation energy for crystallization is extracted from the slope of the relevant plots.

Fig. 6a and b represents the Marotta and the Ozawa-based plots by which the Avrami exponents were derived. The kinetic measurements have been collected in Table 3. Pearson's correlation coefficient (r) was used to determine the correlation between experimental and calculated values about the activation energy of crystallization and Avrami exponent. The correlation coefficient data given in Table 3 were very close to one; it can be seen that a good correlation and strong linear relationship exist between experimental and calculated values.

Obviously, the activation energy for crystallization of the glass $G_{\text{Fe-5}}$ has been considerably decreased compared to that of the glass G_{base} . Furthermore, the Avrami exponent has been changed from approximately 1 for the G_{base} to approximately 2 for the glass $G_{\text{Fe-5}}$. The obtained n near 1 for the glass G_{base} means its surface crystallization tendency. However, $n = 2$ for the glass $G_{\text{Fe-5}}$ reveals its bulk crystallization mechanism with one dimensional crystal growth. Therefore, it can be concluded that Fe_2O_3 has improved the crystallization tendency and changed its mechanism from surface to bulk, showing the nucleating role of Fe_2O_3 in the glass, probably via performing a liquid–liquid phase separation.

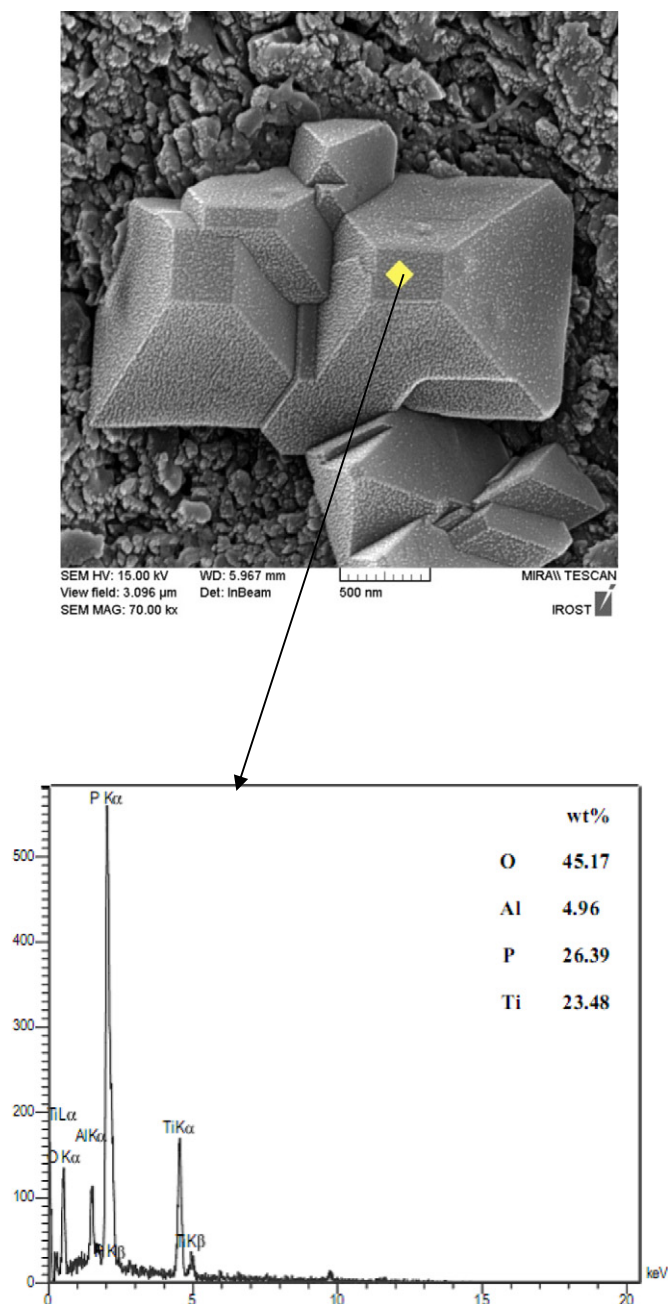


Fig. 8. EDAX pattern taken from the selected area of pyramid crystal.

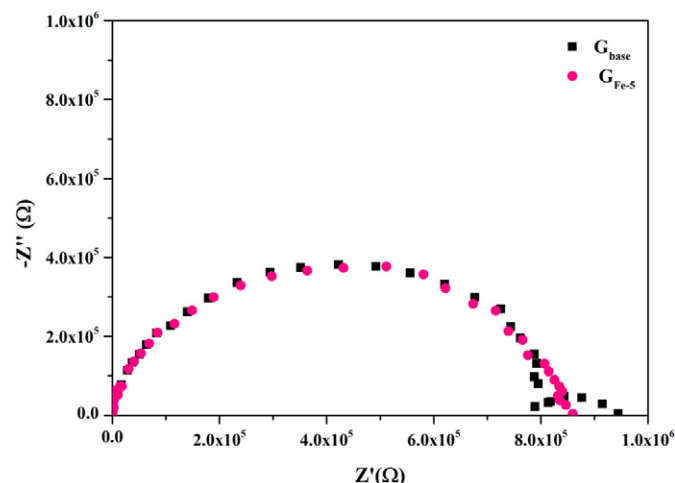


Fig. 9. Room temperature complex impedance diagrams for G_{base} and $G_{\text{Fe-5}}$.

4.5. Evaluation of the microstructure

Fig. 7a, b, c, and d indicates the SEM micrographs of the resulted glass-ceramic G_{base} and $G_{\text{Fe-5}}$ heat treated at 636 °C and 607 °C, respectively for 5 min. Two morphologies of homogeneous near spherical as well as coarse pyramid type are evident in both crystallized specimens. Considering the dominant presence of lithium titanium phosphate (NASICON) in both samples brings to mind that the major fine spherical crystals are related to the NASICON phase.

To identify the inheritance of pyramid-type crystals, EDAX analysis was taken from these crystals according to Fig. 8. The presence of Ti in the EDAX spectrum reveals that these crystals are also referred to the NASICON phase. The pyramid type morphology of the NASICON has been reported previously [43]. To elucidate the formation of the NASICON phase with two morphologies, it should be noted that by decreasing the crystal size of NASICON, morphology change from polyhedron to spherical or inverse has also been reported [43]. Based on the SEM micrographs, the crystal size and distribution in the studied glass-ceramic are somehow different. Due to the lower crystallization temperature and higher tendency towards crystallization, it seems that the glass-ceramic $G_{\text{Fe-5}}$ has been crystallized to a higher degree; the crystal size has been smaller.

4.6. Electrical conductivity measurements

Fig. 9 shows the $Z''-Z'$ plot of the specimens G_{base} and $G_{\text{Fe-5}}$ at 25 °C. A typical semicircle response in the high frequency range is observed in this plot. The conductivity was derived from resistivity values obtained through semicircle's diameter from the intercept with real axis. It is observed that the total resistance (R_{total}) and bulk resistance (R_{bulk}) of $G_{\text{Fe-5}}$ are less than those of the G_{base} . A bulk conductivity of 1.03×10^{-3} and 1.38×10^{-3} S/cm was obtained for the G_{base} and $G_{\text{Fe-5}}$ glass-ceramics, respectively. Based on the obtained XRD results, higher amounts of the NASICON phase in $G_{\text{Fe-5}}$ should be responsible for the improved ionic conductivity of it. The role of Fe_2O_3 in ionic conductivity of the relevant glass-ceramics will be presented in our next work.

5. Conclusion

The addition of Fe_2O_3 to the ternary lithium titanium phosphate glass did not change the kind of the precipitated crystalline phases ($\text{LiTi}_2(\text{PO}_4)_3$ and Li_3PO_4). However, the activation energy for crystallization of the optimized Fe_2O_3 bearing glass ($G_{\text{Fe-5}}$) was considerably less than that of Fe_2O_3 free glass (G_{base}). The Avrami exponent was also changed from near 1 to near 2 confirming the improved crystallization tendency of the glass-ceramic $G_{\text{Fe-5}}$. Based on the FTIR spectra, Fe_2O_3 acts as a modifier in the glass structure and weakens the structural bond.

The presence of the NASICON crystalline phase was evident with two different morphologies of near spherical and pyramid type in the glass-ceramic specimens. The increased amount of the NASICON phase in the glass-ceramic $G_{\text{Fe-5}}$ was responsible for its better ionic conductivity relative to the G_{base} .

References

- [1] H. Hosono, Y. Abe, Fast lithium conducting glass-ceramic in the $\text{CaO}-\text{Li}_2\text{O}-\text{TiO}_2-\text{Al}_2\text{O}_3-\text{P}_2\text{O}_5$ system, *Solid State Ionics* 44 (1991) 293–297.
- [2] H. Hosono, Y. Abe, Porous glass-ceramics with skeleton of the fast lithium conducting crystal $\text{Li}_{1+x}\text{Ti}_2-x\text{Al}_x(\text{PO}_4)_3$, *J. Am. Ceram. Soc.* 75 (1992) 2862–2864.
- [3] B.K. Money, K. Hariharan, Lithium ion conduction in lithium metaphosphate based systems, *Appl. Phys. A* 88 (2007) 647–652.
- [4] X. Xu, Z. Wen, X. Yang, J. Zhang, Z. Gu, High lithium ion conductivity glass-ceramics in $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{TiO}_2-\text{P}_2\text{O}_5$ from nanoscaled glassy powders by mechanical milling, *Solid State Ionics* 177 (2006) 2611–2615.
- [5] R.C. Lucacel, M. Maier, V. Simon, Structural and in vitro characterization of $\text{TiO}_2-\text{CaO}-\text{P}_2\text{O}_5$ bioglasses, *J. Non-Cryst. Solids* 356 (2010) 2869–2874.
- [6] N.A. Ghoneim, A.M. Abdelghany, S.M. Abo-Naf, F.A. Moustafa, K.H. Elbadry, Spectroscopic studies of lithium phosphate, lead phosphate and zinc phosphate glasses containing TiO_2 : effect of gamma irradiation, *J. Mol. Struct.* 1035 (2013) 209–217.
- [7] H.A. ElBatal, A.M. Abdelghany, F.H. ElBatal, K.H. Elbadry, F.A. Moustafa, UV-visible and infrared absorption spectra of gamma irradiated CuO-doped lithium phosphate, lead phosphate and zinc phosphate glasses: a comparative study, *Physica B* 406 (2011) 3694–3703.
- [8] Y. Daiko, H. Yajima, T. Kasuga, Preparation of porous titanium phosphate glass-ceramics for NH_3 gas adsorption with self-cleaning ability, *J. Eur. Ceram. Soc.* 28 (2008) 267–270.
- [9] S.V. Dorozhkin, Amorphous calcium (ortho)phosphates, *Acta Biomater.* 6 (2010) 4457–4475.
- [10] J. Fu, Super conductivity of glass-ceramics in the system $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{TiO}_2-\text{P}_2\text{O}_5$, *Solid State Ionics* 96 (1997) 195–200.
- [11] M.V.N.V.D. sharma, A.V. Sarma, R. Balaji Rao, Electrical conductivity, relaxation, and scaling analysis studies of lithium aluminophosphate glass ceramics, *J. Mater. Sci.* 44 (2009) 5557–5562.
- [12] X. Xu, Z. Wen, Z. Gu, X. Xu, Z. Lin, Lithium ion conductive glass ceramics in the system $\text{Li}_{1+x}\text{Al}_0(\text{Ge}_{1-x}\text{Ti}_x)_{1.6}(\text{PO}_4)_3$, *Solid State Ionics* 171 (2004) 207–213.
- [13] M. Tatsumisago, A. Hayashi, Preparation of lithium ion conducting glasses and glass-ceramics for all-solid-state batteries, *J. Non-Cryst. Solids* 354 (2008) 1411–1417.
- [14] A. Kishioka, Glass formation in the $\text{Li}_2\text{O}-\text{TiO}_2-\text{P}_2\text{O}_5$, $\text{MgO}-\text{TiO}_2-\text{P}_2\text{O}_5$ and $\text{CaO}-\text{TiO}_2-\text{P}_2\text{O}_5$ systems, *Bull. Chem. Soc. Jpn.* 51 (9) (1978) 2559–2561.
- [15] H. Hosono, F. Tsuchitani, K. Imai, Y. Abe, Porous glass-ceramics cation exchangers: cation exchange properties of porous glass-ceramics with skeleton of fast Li ion-conducting $\text{LiTi}_2(\text{PO}_4)_3$ crystal, *J. Mater. Res.* 9 (3) (1994) 755–761.
- [16] K. Yamamoto, Y. Abe, Microporous glass-ceramics in NASICON-type copper (II) titanium phosphate, *J. Am. Ceram. Soc.* 81 (8) (1998) 2201–2204.
- [17] K. Yamamoto, Y. Abe, Enhanced catalytic activity of microporous glass-ceramics with a skeleton of NASICON-type copper (I) titanium phosphate crystal, *Mater. Res. Bull.* 35 (2) (2000) 211–216.
- [18] J. Fu, Fast Li^+ ion conduction in $\text{Li}_2\text{O}-(\text{Al}_2\text{O}_3-\text{Ga}_2\text{O}_3)-\text{TiO}_2-\text{P}_2\text{O}_5$ glass-ceramics, *J. Mater. Sci.* 33 (1998) 1549–1553.
- [19] K. Takada, M. Tansho, I. Yanase, T. Inada, A. Kajiyama, M. Kouguchi, S. Kondo, M. Watanabe, Lithium ion conduction in $\text{LiTi}_2(\text{PO}_4)_3$, *Solid State Ionics* 139 (2001) 241–247.
- [20] N. Anantharamulu, K. Koteswara Rao, G. Rambabu, B. Vijaya Kumar, V. Radha, M. Vithal, A wide-ranging review on Nasicon type materials, *J. Mater. Sci.* 46 (2011) 2821–2837.
- [21] T. Ozawa, A new method of analyzing thermogravimetric data, *Bull. Chem. Soc. Jpn.* 38 (1965) 1881–1886.
- [22] H.E. Kissinger, Variation of peak temperature with heating rate in differential thermal analysis, *J. Res. Natl. Bur. Stand.* 57 (4) (1956) 217–221.
- [23] J.E. Augis, J. Bennett, Calculation of the avrami parameters for heterogeneous solid state reactions using a modified the Kissinger method, *J. Therm. Anal.* 13 (1978) 283–292.
- [24] F. Branda, A. Buri, A. Marotta, S. Saiello, Kinetic of crystal growth in $\text{Na}_2\text{O}-2\text{SiO}_2$ glass: a DTA study, *Thermochim. Acta* 77 (1–3) (1984) 13–18.
- [25] I. Dyament, E. Korin, J. Hormadaly, Non-isothermal crystallization kinetics of $\text{La}_2\text{CaB}_6\text{O}_{19}$ from glass, *J. Non-Cryst. Solids* 357 (7) (2011) 1690–1695.
- [26] M. Erol, S. Kucukbayrak, A. Ersoy-Mericboyu, Influence of particle size on the crystallization kinetics of glasses produced from waste materials, *J. Non-Cryst. Solids* 357 (1) (2011) 211–219.
- [27] Mousa M.A. Imran, Crystallization kinetics, glass transition kinetics, and thermal stability of $\text{Se}_{70-x}\text{Ga}_{30}\text{In}_x$ ($x = 5, 10, 15$, and 20) semiconducting glasses, *Physica B* 406 (3) (2011) 482–487.
- [28] S. Mahadevan, A. Giridhar, A.K. Singh, Calorimetric measurements on As–Sb–Se glasses, *J. Non-Cryst. Solids* 88 (1986) 11–34.
- [29] M. Saad, M. Poulin, Glass forming ability criteria, *Mater. Sci.* 19&20 (1987) 11.
- [30] M. Avrami, Kinetics of phase change, *J. Chem. Phys.* 7 (1939) 1103 (8 (1940) 212, 9 (1941) 177).
- [31] W.A. Johnson, K.F. Mehl, Reaction kinetics in process of nucleation and growth, *Trans. Am. Inst. Min. Metall. Pet. Eng.* 135 (1939) 416.
- [32] S.T. Reis, M. Karabulut, D.E. Day, Chemical durability and structure of zinc-iron phosphate glasses, *J. Non-Cryst. Solids* 292 (2001) 150–157.
- [33] N. Shibata, M. Horiguchi, T. Eda, Raman spectra of binary high-silica glasses and fibers containing GeO_2 , P_2O_5 and B_2O_3 , *J. Non-Cryst. Solids* 45 (1981) 115–126.
- [34] A.M.B. Silva, R.N. Correia, J.M.M. Oliveira, M.H.V. Fernandes, Structural characterization of $\text{TiO}_2-\text{P}_2\text{O}_5-\text{CaO}$ glasses by spectroscopy, *J. Eur. Ceram. Soc.* 30 (2010) 1253–1258.
- [35] A. Shaim, M. Et-tabirou, Role of titanium in sodium titanophosphate glasses and a model of structural units, *Mater. Chem. Phys.* 80 (2003) 63–67.
- [36] G.M. Krishna, N. Veeraiiah, N. Venkatramiah, R. Venkatesan, Induced crystallization and physical properties of $\text{Li}_2\text{O}-\text{CaF}_2-\text{P}_2\text{O}_5:\text{TiO}_2$ glass system. Part I. Characterization, spectroscopic and elastic properties, *J. Alloys Compd.* 450 (2008) 477–485.
- [37] T.Y. Wei, Y. Hu, L.G. Hwa, Structure and elastic properties of low-temperature sealing phosphate glasses, *J. Non-Cryst. Solids* 288 (2001) 140–147.
- [38] Y.M. Moustafa, K. El-Egili, Infrared spectra of sodium phosphate glasses, *J. Non-Cryst. Solids* 240 (1998) 144–153.
- [39] S.T. Reis, D.L.A. Faria, J.R. Martinelli, W.M. Pontuschka, D.E. Day, C.S.M. Partiti, Structural features of lead iron phosphate glasses, *J. Non-Cryst. Solids* 304 (2002) 188–194.
- [40] F.H. ElBatal, Y.M. Hamdy, S.Y. Marzouk, UV-visible and infrared absorption spectra of transition metals-doped lead phosphate glasses and the effect of gamma irradiation, *J. Non-Cryst. Solids* 355 (2009) 2439–2447.
- [41] R.K. Brow, D.R. Tallant, S.T. Myers, C.C. Philfer, The short-range structure of zinc polyphosphate glass, *J. Non-Cryst. Solids* 191 (1995) 45–55.
- [42] K. Meyer, Characterization of the structure of binary zinc ultraphosphate glasses by infrared and Raman spectroscopy, *J. Non-Cryst. Solids* 209 (1997) 227–239.
- [43] Ch. Myung Chang, Y. Lee, S. Hyeon Hong, Spark plasma sintering of $\text{LiTi}_2(\text{PO}_4)_3$ -based solid electrolytes, *J. Am. Ceram. Soc.* 88 (7) (2005) 1803–1807.